

Uttarakhand State Geo-Portal for Decentralized Governance



J. S. Rawat and R. Dobhal

1 Introduction

In order to fulfil the objectives of Digital India Initiative of the Government, the present “e-governance” system is gradually being transformed into “g-governance” to improve transparency and decentralize the process of governance to lower area levels. In the recent years, there has been an increased need for more timely access to maps to facilitate effective planning and administration. In e-governance, we are getting information quickly through computers in the form of tables, text and analogue maps but in the g-governance system, one can access quickly digital or geo-stamped map showing geographic location and information of the targeted area/place for decision support. Thus, g-governance makes use of concepts and principles from the domain of Geographic Information Science [1] by utilizing IT, MIS and Communication Technologies. For developing the information system for g-governance, an enormous amount of geo-stamped data gathering, compilation, analysis and manipulation through the geospatial technologies is needed. The next decade is likely to witness complete switch over to g-governance in India with all the office records and maps accessible to planners and decision-makers in digital form. Under the Natural Resources Data Management System (NRDMS) Programme of the Department of Science and Technology, Government of India (DST, GOI), efforts are being made to promote geospatial technologies in the country by establishing about 72 district-level GIS Centres spread in different States of the country towards establishing State Spatial Data Infrastructures (SDI) for supporting implementation of the g-governance concept. Uttarakhand is one of the few states where all the districts

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have been equipped with district-level GIS Cells to support data management by conducting GIS training programmes for the staff and personnel of the line departments. One District GIS Cell initially set up in 2009 at the beginning of the DST Initiative in Almora district of the state and the co-located Centre for NRDMS established at the Department of Geography, Kumaun University, Almora have significantly contributed to this process of change in state-level geospatial data management towards the implementation of g-governance.

2 Background

2.1 District GIS Cell at Almora

In an earlier pilot phase of the related activities, through a research project funded by the DST GOI, a District Digital Data Base Centre Almora was established at the Department of Geography in the Kumaun University in 2002. At that time the Centre was the 54th district database centre of the country. Through this project the GIS infrastructure of Almora District was developed in association with the district line departments as per their respective geographic information needs. During this second phase (2006–2008) the GIS work of district Almora was updated and upgraded in web-accessible mode and developed and demonstrated as a template of a District Geo-Portal that could be successfully linked to the national level India Geo-Portal hosted at the NSDI, New Delhi. In the third phase beginning in 2008, considering the requirements of further technology customisation & development and intensive capacity building among the staff and personnel in Districts on Geospatial Technologies, a Centre of Excellence for NRDMS in Uttarakhand (COE NRDMS) was set up in 2009.

2.2 Centre for NRDMS at Kumaun University

With the mission to build professional education and applied research capacity in the latest geospatial technologies for various stakeholders, viz., academia, industry, governance and the civil society with special reference to the Uttarakhand State, the COE NRDMS has the following objectives:

- (i) conduct education and research programmes in geospatial technologies, viz., Remote Sensing and Geographic Information Science;
- (ii) develop spatial data infrastructure and GIS of the Uttarakhand state as per needs of the government departments; and
- (iii) need-based capacity building of various stakeholders, viz. academia, industry, governance and civil society in application of geospatial technologies.

To replicate the Web-GIS template of District Almora in all the Districts of the State, a project on “Development of Standards-Based Uttarakhand Geo-Portal for Decentralised Governance” funded jointly by the DST, GOI and the DST, Government of Uttarakhand on 70% and 30% cost-sharing basis and collaboratively launched by the Uttarakhand State Council of Science and Technology (UCOST), Dehradun, and the COE NRDMs, Department of Geography, Kumaun University, Almora in November 2012. The project has ended in March 2018. With the help of this joint collaborative research project, Uttarakhand Geo-Portal has been developed for its application in planning and decision-making by the end-users. The specific characteristics of the Geo-Portal include the following:

- (i) it is developed as per the needs of the users, i.e. different line departments;
- (ii) it is developed in association with the line departments while sensitizing them on its utility and building their technical capacities;
- (iii) it has provision of regular updating through the line departments;
- (iv) it has a regular management system comprising of technically trained personnel at State, District and District line department levels; and
- (v) it has also provision for developing its applications and upgradation as per the needs of the line departments.

The objective of this paper has been to provide the details of the method adopted in State-level spatial data management through the State Geo-Portal in the framework of the State SDI, its management structure, the architecture utilized in its implementation, the conceptual data model underlying the geospatial database that remains at its heart, provision of the standards-based data services for state-wide access by the end-users. Assessment of the geographic information needs of the end-users is vital to the development of this infrastructure to ensure better utilization.

3 End-User Need Assessment

The first two-day workshop on Information Need Assessment for Land and Water Management in District Almora was organized in May 2000 District planners/administrators/representatives of Municipal Board and Scientists of GB Pant Institute of Himalayan Environment and Development participated. Through this workshop, department-wise documentation of digital data needs for governance were assessed [2]. Based on these end-user information needs, the proposal was developed in association with the level planners and administrators for setting up a data management centre at Almora in 2002 and was supported by DST during 2002–2005.

To update the information needs for district planning and decision-making, the second workshop in the State on end-users need assessment was organized in August 2012 at Administrative Training Institution Nainital. In this workshop, the heads of line departments and administrators of Nainital district, participated and an updated digital information need of end-users was documented [3]. Based on these two end-users information need assessments, the district-wise GIS infrastructure of different

line departments was developed in the state in association with the end-users by establishing District GIS Cell in each district.

4 Uttarakhand (UKH) Geo-Portal

4.1 Definition and Objectives

The UKH Geo-Portal is an Internet-based directory of geospatial data, metadata, and services for the State that allows data providers and users to share and explore geospatial information related to boundaries, demography, agro- and socio-economy, resources, infrastructure facilities with attributes. Geospatial data sets generated, maintained and provided by various concerned departments of the State Government, academia, business/industry or non-governmental organizations (NGOs) of Uttarakhand are proposed to be made accessible through a clearing house mechanism. To develop and maintain/update the data sets on Uttarakhand Geo-Portal, District GIS Cells have been established in all the districts of the State. Towards this end, Government of Uttarakhand has appointed a State-level Nodal Officer in each Department and a District-level Nodal Officer in each district of the State. For better coordination and management of data-related activities towards the development of the UKH SDI, the State Government has constituted a State SDI Steering Committee and a State SDI Executive Committee.

Thus, through the UKH Geo-Portal, a common platform of Web GIS at 1:50,000 scale has been developed for better managing all the spatial data sets of the State so that those could be updated and made sharable for use in GIS packages all over the State. Following the specifications of OGC's Geospatial Portal Reference Architecture, Ukh Geo-Portal development and updating has been in progress. It has provision to provide portal services, portrayal services, data services and catalogue services. The portal services shall be available to address the needs of the viewer clients, discovery clients, management clients, authentication and access control and exposed services. The portrayal services include map services, styling services, coverage services and the map context services. The data services include features, coverage and symbology management. The catalogue services include data discovery, service discovery, catalogue update and query language. The development which is still going on is being carried out by the users themselves with the technical support of GIS Cell established in each district/COE NRDMS Centre of Kumaun University.

The Geo-Portal is essentially being used for the following objectives:

1. Bringing together GIS-based knowledge and information of the Uttarakhand State under a common platform.
2. Providing a common gateway for various data generating agencies to share the information across various government departments, academia, business/industry and NGOs.
3. Making accessible a geographic data dictionary and map directory for the State.

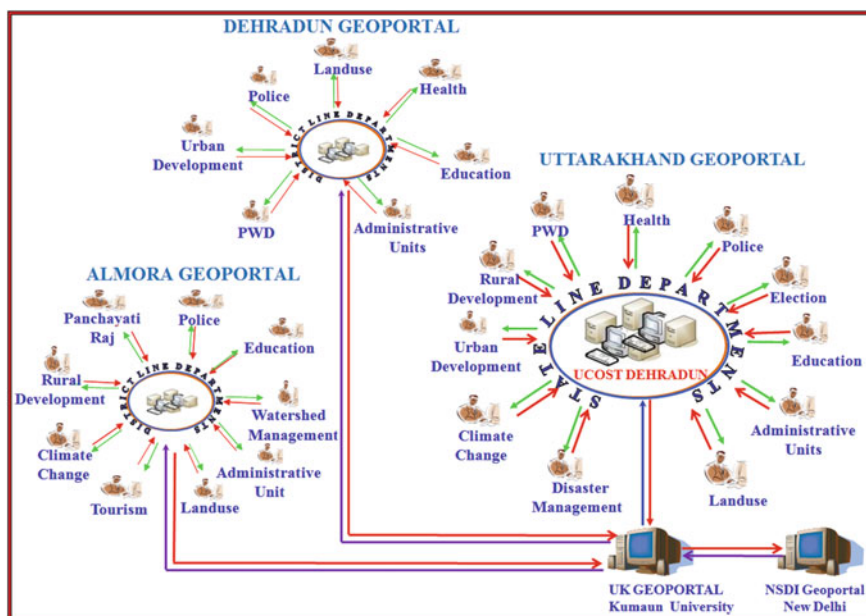


Fig. 1 High-level Architecture of Uttarakhand Geo-Portal for improving access to geospatial data at national, state and district levels

4. Analysing the needs of various government schemes/programmes.
5. Facilitating development of decision support systems and helping in local-level planning.

Figure 1 presents the architecture of the UKH Geo-Portal which reveals that it is being linked with all the district GIS Cells and also tagged with the India Geo-Portal.

4.2 Structure and Responsibility

The UKH Geo-Portal has seven major organizational components. These are

- (i) Department of Planning, Government of Uttarakhand;
- (ii) UCOST Dehradun;
- (iii) NRDMS Centre, Department of Geography, Kumaun University;
- (iv) Uttarakhand Space Application Centre (USAC) Dehradun;
- (v) District GIS Cells;
- (vi) District line departments; and
- (vii) State line departments.

5 Capacity Building

For building the capacity of the State in development and application of GIS in decentralised and good governance, sector-wise training workshops have been organized. For example, Public Works Department (PWD) for development of GIS of Roads [4], Department of Revenue for modernization of land records [5], and Department of Police for police administration [6, 7], etc.

5.1 *Setting-Up District GIS Cells*

For developing GIS with the involvement of the end-users, i.e. line departments, the State Government has made provision of setting-up GIS Cell at each district head-quarter, under the control of district magistrates. By conducting training workshops jointly by the COE NRDMS and the UCOST Dehradun on “GIS Development and Setting-up District GIS Cell” [8] at the headquarter of each district of the state, the GIS Cells have been established in all the 13 districts of Uttarakhand. The responsibility of the district GIS Cells has been to develop and regularly update GIS of different line departments and to build their technical capacity in application of GIS in governance.

5.2 *Compilation and Publication of Metadata*

A provision has been made in the system to permit various GIS data providers in the State to publish their metadata sets into the Geo-Portal/centralized database and get them registered directly at UCOST as per the metadata standards prescribed in BIS 16439. For data updation and sharing of up-to-date data through geo-web services, a metadata format as per the standard has been prepared for each layer of information.

6 Geographical Data Modelling

In geographic information systems, a data model is a mathematical construct for representing geographical objects in a formal and standardized way. In the design of the database for the UKH Geo-Portal, data models have been developed in various sectors like boundaries, physiography, hydrography, land use, geology, soils. Figure 2 provides overviews of the sector-wise data models used in the UKH Geo-Portal covering the end-user needs of 52 different sectors of the State and detailed feature-level classes in UML for the State Police Department have been shown in Fig. 3.

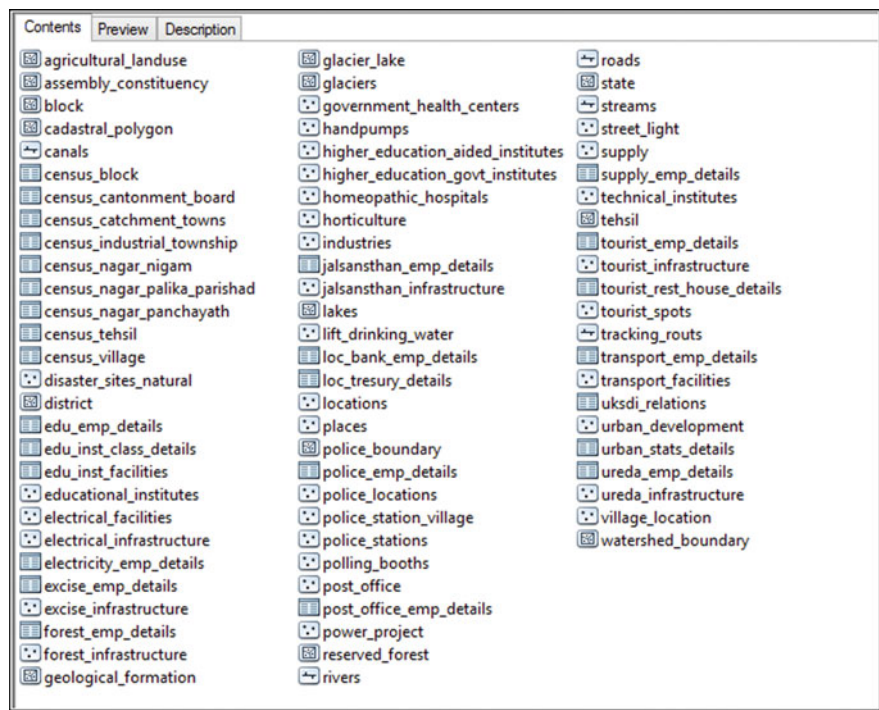


Fig. 2 Selected features from 52 different sectoral data layers of UKH Geo-Portal

6.1 Development of GIS Database

Based on the above data models, the GIS database of the State has been developed in association with district line departments and the district GIS Cells The database consists of sectoral data sets re-engineered in line with the Conceptual Schema Language specifications of the International Standardisation Organisation (ISO 19103) and implemented in the open source PostGreSQL/PostGIS. UCOST Dehradun hosts the UKH Geoportal along with the database.

6.2 Development of Web-GIS

To develop and demonstrate methods and techniques for better managing and utilising geospatial data sets in Uttarakhand State by the line departments, efforts are being made to make the database accessible along with Web GIS functionalities for processing. The spatial data sets in the Geo-Portal are also accessible in ESRI’s shape format for the benefit of the ARC GIS users in the State. Associated metadata for the GIS data sets are appropriately included in the database to support discovery and

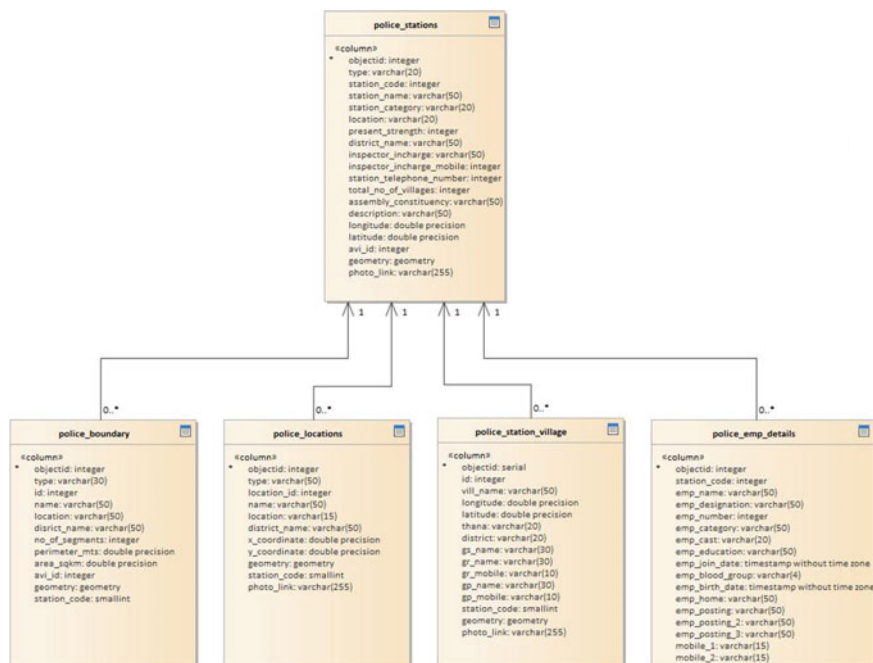


Fig. 3 UML Class diagram of different data items of the Department of Police, Government of Uttarakhand

access by concurrent users over the net. Mechanism for updation of existing data sets on the database using OGC standards-based services has been developed and implemented. Web Feature Service (WFS) and Web Map Service (WMS) have been provided to facilitate updation/maintenance of the database from users located in the individual line departments/agencies of the Government of Uttarakhand. In due course, depending on requirement, individual line departments/agencies of the government may be persuaded to develop and maintain their spatial/attribute data nodes as per their mandates in the interest of users/stakeholders of local-level planning for faster access to up-to-date data.

6.3 UKH Geo-Portal Servers

The UKH Geo-Portal hosted at the UCOST Dehradun consists of three servers—the database server, security server and the application server.

6.4 Security System

In order to secure the geo-portal/clearing house from unauthorized access, an appropriate security software module having provisions to safeguard data sets, manage user identity and transaction security; detect unauthorized intrusion, permit authorization/authentication, and non-repudiation-based access controls on resources have been installed. The security system has facilities to create federated access and single sign-on, on-the-fly creation of secured access and functionalities for user-friendly operations and access to data services from the Geo-Portal.

7 Provision of Standards-Based Data Services

UKH Geo-Portal provides services adhering to the ISO/OGC specifications like WMS, WFS, WCS, CS-W, WFS-T, and WPS.

A Web Map Service (WMS) is a standard protocol for serving georeferenced maps (spatial data with colours, styles, legend, annotation, etc.) for visualization of data sets by the end-users while taking decisions. Web Feature Service (WFS) provides an interface allowing requests for geographical features across the Web using platform-independent calls. WFS is the provision for upload/download of OGC's Geography Markup Language (GML) data (in GML 2.1.2 specification) for further processing in GIS package subject to authorization for download policy for sharing and download of geographical feature data by authorized user is currently being framed for approval by the State Cabinet. **Web Coverage Service Interface Standard (WCS)** provides an interface allowing requests for geographical coverage including map photographs or satellite images. **Web Processing Service (WPS)** is designed to standardize the way GIS calculations are made accessible to the end-users over the net. WPS can describe any calculation (i.e. process) including all of its inputs and outputs, and trigger its execution as a web service. **Catalogue Services on Web (CSW)** defines common interfaces to discover, browse and query metadata about data, services and other potential resources on UKH Geo-Portal's local register. The Uttarakhand local registers are proposed to be in turn linked to the proposed National Data Registry of NSDI for automatic harvesting and updation of the national registers for countrywide access and utilization.

8 Data Policy

The equitable, productive and sustainable developmental planning as well as good and inclusive governance need quality and authentic updated data and information. Different stakeholders of the State Government have been generating large quantities of different forms of data and information with public funds. But this activity,

most of the times, creates redundant data generated utilizing public money, time and human resources because of lack of information about such data and openness in sharing the data sets between different government departments and stakeholders. General public also seems to be deprived of searching, discovering and accessing government-owned data from the concerned departments for participation in the governance process although most of such data may be non-sensitive in nature and shareable, because State Government departments do not generally publish catalogue or metadata of available data or even the data themselves. Sharing of relevant data is however essential to support effective and transparent governance. Further, due to lack of proper standards and interoperability of the data and information, inter-departmental sharing and integration of such data and information become difficult and ineffective, apart from its access and use by common citizens.

Hence, there is a need to evolve a protocol to facilitate easy accessibility, effective sharing and efficient utilization of the large amount of data generated and stored in disparate mode among the different entities of the State Government. There is a need for both independent dedicated data enterprise system in different government departments and need to share critical and relevant information across the departments with a central repository to make integrated data set available for informed decision-making. This calls for adoption of a data policy in the state which conceptualizes provisions for proper data management methods, defining data standards and processes for providing proactive and open access to such data.

The Draft State Data Policy has been formulated with a mandate of using government-owned data and information in the interest of concern citizens of the State. Under this policy, line departments will be mandated to share data and information with each other and other stakeholders including common citizen thus enhancing sustainable and productive use of the data sets in planning process of the State. The State Data Policy will provide greater scope for unified decision-making in the State. The basic objectives of Uttarakhand State Data Policy are:

- (i) to support generation and storage of data and information in all forms at all State Government departments in a standard and interoperable format;
- (ii) to implement a common protocol of updated data access and sharing in an integrated, proactive and periodic manner;
- (iii) to promote use of all data and information including standardized geospatial data sets in planning, implementation and monitoring developmental programmes, disaster mitigation and scientific research; and
- (iv) to facilitate accessibility of authentic data and information to by all the stakeholders through an electronic network within the framework of various related Policies, Acts and Rules of Government of Uttarakhand and Government of India made from time to time with a view to empowering the citizens.

Benefits of Uttarakhand State Data Policy are expected as follows:

- (i) Avoiding data redundancy: By sharing data the practice of different State Government department/organizations collecting the same data will be avoided, resulting in significant cost and time savings.

- (ii) Open information on data ownership: The preparation of data catalogue/metadata (data about data) by each government department will identify the ownership for the principal data sets to users and enable information about available data sets.
- (iii) Maximizing use: Ready access to government-owned data using valuable public resource will enable more extensive use of data generated for the benefit of the citizens of the State.
- (iv) Maximized integration: By adopting common standards for creation, generation, storage and transformation, the data can be made interoperable making integration of individual data sets feasible for holistic planning.
- (v) Decision support systems generation: Interoperable data and information will support preparation of effective decision support systems needed in development planning, managing assets, improving socio-economic conditions, mitigating disasters, and maintaining internal security of the State, etc.
- (vi) Equity of access: A more open and transparent data policy will ensure better access to data and information by all bonafide users in the State.

Using this policy, Government of Uttarakhand will strive to form one umbrella under which users can be able to access all data and information as per their choices. The scope and objectives of this policy may be expanded and enriched in future, as the data management technologies and procedures are change day by day.

For developing data policy and mechanism for data sharing among the users within and outside the State a Committee has been constituted under the Chairmanship of the Secretary, Science and Technology, Government of Uttarakhand.

9 Future Works

Thus, through the UKH Geo-Portal, a common platform of Web GIS at 1:50,000 scale has been developed for better managing all the spatial data sets of the State so that those could be updated and made sharable for use in GIS packages all over the State. The Uttarakhand State also has a coordinating structure with nodal officers at state/district levels. Now a road map for Uttarakhand State Geospatial Data Assets Management is a prerequisite and is proposed to be approved by the State Government for implementation by all the GIS users in the State. A data management (updatation/maintenance/sharing) plan should be developed and implemented by UCOST, State Line Department Nodal Officers through a series of workshops. Regular workshops (3–4 per year) at State and District levels need to be organized for reviewing the progress of the management plan. The UKH Geo-Portal database needs to be continuously improved with not only data updatation but with high-resolution data addition for its upgradation depending on GIS applications for different projects.

In view of the above necessity of smooth functioning and sustenance of the UKH Geo-Portal in order to provide uninterrupted access to data sets by end users, there is an urgent of need for development of a road map for formulation and publica-

tion of a framework and detailed guidelines for spatial data assets management and applications for the use of state.

10 Conclusions and Recommendation

Geo-Portals are becoming a major resource for gaining quicker access to geospatial data and services useful in decentralising governance on a Statewide scale. They are also contributing to sound decision-making, effectively implementing e-government applications and provision of better services to citizens. Geo-Portals are getting increasingly necessary for emergency preparedness and response, counterterrorism, monitoring of and response to pandemics, and environmental protection. In order for these transnational applications to be effective, Geo-Portal must interoperate. The interoperability can be achieved only through consistent and structured implementation of interface and encoding standards. The UKH Geo-Portal has successfully demonstrated that by building capacity of the end-users, i.e. line departments, they can convert their office records/data/document and maps into GIS environment and can also update GIS on regular basis. It is recommended that GIS should be developed in the country by capacity building of the end-users by which they could develop GIS as per their own need and could make application of GIS in governance, i.e. in planning, implementation, monitoring, management, surveying and research, and in administration for bringing transparent, good and decentralized governance by which the country could reach up to the logical culmination of the Digital India Mission.

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